1 Randomized Covariance Transport

This chapter formulates the climate-facing problem in the thesis that is closest to its numerical core. The climate-model application matters, but the main mathematical object is not the full post-processing workflow. It is a regularized dependence-correction map built from empirical covariance operators after marginal adjustment. This viewpoint keeps the discussion anchored in randomized numerical linear algebra and mixed-precision perturbation analysis while still preserving a clear connection to multivariate bias correction.

Existing multivariate bias-correction methods already show why dependence structure matters. Quantile-delta mapping (QDM) corrects marginal distributions while preserving modeled changes in quantiles [1]. Methods such as MBCp and MBCr combine marginal correction with dependence adjustment, and MBCn and later rank-based approaches push further toward correcting the full multivariate distribution [2] [3] [4] [5]. At the same time, the climate literature repeatedly warns that bias correction should not be treated as a black-box distributional repair, especially when the method modifies structures that matter dynamically or physically [6]. For the present thesis, this leads to a narrower and more analyzable question: can one replace an expensive dependence-correction step by a randomized low-rank approximation whose finite-precision behavior can still be controlled?

The main claim of this chapter is therefore methodological. Rather than analyze the full nonlinear climate workflow at once, we isolate a covariance-based transport problem in a latent dependence space. This reduction is still rich enough to capture the main tradeoffs between sketch dimension, regularization, conditioning, and arithmetic precision, but it is much cleaner than attempting a direct perturbation analysis of a fully nonlinear multivariate adjustment scheme.

1.1 Latent Dependence Representation

Let

$$X_h \in \mathbb{R}^{n\times p}, \qquad X_f \in \mathbb{R}^{m\times p}, \qquad Y_h \in \mathbb{R}^{n\times p}$$

denote historical climate-model data, future climate-model data, and historical reference data, respectively. The column dimension p may encode several physical variables, nearby locations, lags, seasonal features, or any local multivariate state used for downstream post-processing.

The first stage remains standard. Each marginal series is corrected by a univariate method of QDM type,

$$\widetilde{X}_h, \widetilde{X}_f = \mathrm{QDM}(X_h, X_f; Y_h),$$

so that the eventual output inherits the desired marginal behavior from the reference historical data together with modeled changes in quantiles [1]. What remains is the dependence correction.

To express dependence in a form closer to linear algebra, it is convenient to pass to a latent Gaussianized space. The precise standardization map can be chosen in several closely related ways, but it should avoid empirical probabilities equal to 0 or 1, since Φ^{-1} would then produce infinities. A convenient finite definition is obtained from plotting-position ranks. For each column j, write

$$u_{X,ij} = \frac{\operatorname{rank}(\widetilde{X}_h(i,j)) - \frac{1}{2}}{n}, \qquad u_{Y,ij} = \frac{\operatorname{rank}(Y_h(i,j)) - \frac{1}{2}}{n},$$

and similarly

$$u_{f,ij} = \frac{\operatorname{rank}(\widetilde{X}_f(i,j)) - \frac{1}{2}}{m}$$

for the future sample. Ties are handled by average ranks; any fixed deterministic tie-breaking rule would serve equally well. We then define

$$Z_X(i,j) = \Phi^{-1}(u_{X,ij}), \qquad Z_Y(i,j) = \Phi^{-1}(u_{Y,ij}),$$

and similarly obtain Z_f from u_f. In this representation, marginal correction is separated from dependence correction: the marginals are handled before and after the latent transform, while the multivariate adjustment acts on the standardized latent variables.

This reduction is not meant to claim that all relevant climate dependence is Gaussian. It is a modeling step that isolates a dependence operator that is computationally tractable and mathematically interpretable. In particular, it allows the dependence-correction stage to be expressed through covariance operators in a finite-dimensional inner-product space.

1.2 Regularized Covariance Transport

Assume that the columns of Z_X and Z_Y have been centered. Their empirical covariance operators are

$$C_X = \frac{1}{n} Z_X^\top Z_X, \qquad C_Y = \frac{1}{n} Z_Y^\top Z_Y.$$

The dependence-correction step is then represented by the regularized transport map

$$T_\lambda = (C_X + \lambda I)^{-1/2}(C_Y + \lambda I)^{1/2},$$

where $\lambda > 0$ is a regularization parameter. In the language of optimal transport, this is a *regularized whitening–coloring covariance transport*: the map first whitens the source covariance (inverse square root) and then colors the result with the target covariance (square root). It coincides with the Bures–Wasserstein optimal-transport map when C_X and C_Y commute, but in general it is not the symmetric optimal-transport map. The present thesis adopts it primarily for its algebraic tractability and its natural connection to matrix-function perturbation theory, not as a claim of optimality in the transport sense.

The role of λ is both numerical and conceptual. Numerically, it keeps the source and target operators strictly positive definite and prevents the inverse square root from reacting catastrophically to small singular values or nearly redundant variables. Conceptually, it acknowledges that in high-dimensional post-processing tasks one should not insist on recovering dependence structure below the scale at which the empirical covariance is already poorly determined.

The corrected latent future data are then obtained as

$$Z_f^{\mathrm{corr}} = Z_f T_\lambda.$$

Remark 1.1 (Regularization bias)**.** For $\lambda = 0$, the idealized transport map $T = C_X^{-1/2} C_Y^{1/2}$ satisfies

$$T^\top C_X T = C_Y$$

whenever the inverse square root is well defined. For $\lambda > 0$, the regularized map instead satisfies

$$T_\lambda^\top (C_X + \lambda I) T_\lambda = C_Y + \lambda I,$$

and therefore transports the regularized covariance exactly, not the original unregularized covariance. Thus λ controls both numerical stability and modelling bias. ⌟

Finally, because a linear map in latent space will generally disturb the exact marginal distributions, the latent correction is used only to update ranks. For each variable, the values from the QDM-corrected future sample $\widetilde{X}_f(:, j)$ are sorted and then rearranged according to the ranks of $Z_f^{\mathrm{corr}}(:, j)$. The final output therefore combines

$$\text{marginals from QDM} \quad + \quad \text{dependence from covariance transport.}$$

This construction should be viewed as a lower-rank and more analyzable alternative to full multivariate distribution matching. It is less ambitious than MBCn in what it tries to capture, but it is correspondingly better aligned with the perturbation tools of numerical linear algebra.

1.3 Randomized Low-Rank Approximation

The direct formation and decomposition of C_X and C_Y is unattractive when p is large. The thesis therefore proposes to approximate the covariance operators by randomized low-rank surrogates. Let $\Omega \in \mathbb{R}^{p \times \ell}$ be a sketching matrix with $\ell = r + q$, where r is the target rank and q an oversampling parameter. Then the dominant range of C_X may be sampled through

$$Y_X = C_X \Omega = \frac{1}{n} Z_X^\top (Z_X \Omega),$$

and analogously for C_Y. This is precisely the type of factorization-friendly structure for which randomized low-rank approximation is effective [7] [8].

If Q_X is an orthonormal basis for the range of Y_X, the covariance operator is approximated by

$$C_X \approx Q_X B_X Q_X^\top, \qquad B_X = Q_X^\top C_X Q_X,$$

with an analogous approximation for C_Y. The transport map is then built from

$$\widehat{C}_{X,\lambda} = Q_X B_X Q_X^\top + \lambda I, \qquad \widehat{C}_{Y,\lambda} = Q_Y B_Y Q_Y^\top + \lambda I.$$

The computational advantage is immediate: the numerically demanding work is concentrated in matrix products of the form $Z\Omega$ and $Z^\top(Z\Omega)$, while the eigenanalysis is transferred to the small dense matrices B_X and B_Y.

Proposition 1.2 (Randomized covariance approximation)**.** Let $C \in \mathbb{R}^{p\times p}$ be symmetric positive semidefinite, and let Q be produced by a randomized range finder with target rank r and oversampling parameter q. Then the range-finder compression error

$$\|(I - QQ^\top)C\|_2$$

is controlled by the spectral tail of C beyond rank r. In particular, standard range-finder bounds imply that there exist oversampling-dependent constants $c_{r,q}$ and $d_{r,q}$ such that

$$\mathbb{E}\,\|(I - QQ^\top)C\|_2 \le c_{r,q}\,\mu_{r+1}(C) + d_{r,q}\left(\sum_{j>r}\mu_j(C)^2\right)^{1/2},$$

where $\mu_1(C) \ge \mu_2(C) \ge \cdots \ge 0$ are the eigenvalues of C in nonincreasing order Chs. 10–11 [7] Sec. 8 [8].

Proof (Proof sketch)**.** This is the standard randomized range-finder estimate specialized to a symmetric positive semidefinite matrix. Since C is self-adjoint, its singular values are precisely the eigenvalues $\mu_j(C)$. $\square$

It is important to distinguish the range-finder residual from the actual surrogate error. The range finder controls $\|(I - QQ^\top)C\|_2$, which measures how well the column space of Q captures the action of C. The low-rank surrogate actually used in the transport map is $\widehat{C} = QBQ^\top$ with $B = Q^\top CQ$, and its approximation error is

$$\|C - QBQ^\top\|_2 = \|C - QQ^\top CQQ^\top\|_2.$$

For a symmetric positive semidefinite matrix, these two quantities are related but not identical: the range residual upper-bounds the surrogate error because $\|C - QQ^\top CQQ^\top\|_2 \le \|(I - QQ^\top)C\|_2 + \|CQQ^\top - QQ^\top CQQ^\top\|_2$, and the second term vanishes when C commutes with the projection $QQ^\top$. In the experiments, the recorded covariance error is the surrogate error $\|C - \widehat{C}\|_2$, while the theory provides bounds on the range-finder residual. The gap between them is typically small for SPD matrices with rapid spectral decay.

Moreover, the regularized matrix functions of a low-rank surrogate can be applied without forming a full dense $p \times p$ matrix. If $Q \in \mathbb{R}^{p\times r}$ has orthonormal columns and $B \in \mathbb{R}^{r\times r}$ is symmetric positive semidefinite, then

$$(QBQ^\top + \lambda I)^{-1/2} = \lambda^{-1/2} I + Q\Big((B+\lambda I_r)^{-1/2} - \lambda^{-1/2} I_r\Big)Q^\top,$$

and similarly

$$(QBQ^\top + \lambda I)^{1/2} = \lambda^{1/2} I + Q\Big((B+\lambda I_r)^{1/2} - \lambda^{1/2} I_r\Big)Q^\top.$$

These identities show that the transport layer only needs matrix functions of the small core matrix $B + \lambda I_r$, together with applications of Q and $Q^\top$.

The point is not only efficiency. A low-rank approximation introduces an explicit randomized error scale. Once this scale is visible, it becomes meaningful to ask whether lower precision is acceptable because its rounding error remains smaller than the approximation error already introduced by the randomized compression.

1.4 Independent and Shared Reduced Bases

The randomized low-rank approximation described above produces an orthonormal basis Q_X for the dominant range of C_X and a basis Q_Y for the dominant range of C_Y. When the source and target covariance operators are approximated independently, the resulting surrogates take the form

$$C_X \approx Q_X B_X Q_X^\top, \qquad C_Y \approx Q_Y B_Y Q_Y^\top,$$

where $B_X = Q_X^\top C_X Q_X$ and $B_Y = Q_Y^\top C_Y Q_Y$ are small dense core matrices. This independent-basis construction is the natural output of two separate range-finder calls.

The difficulty is that Q_X and Q_Y may span different reduced coordinate systems. Even if each covariance is approximated reasonably well in isolation, the transport map combines the inverse square root of one surrogate with the square root of the other:

$$\widehat{T}_\lambda = (Q_X B_X Q_X^\top + \lambda I)^{-1/2}(Q_Y B_Y Q_Y^\top + \lambda I)^{1/2}.$$

When $Q_X \neq Q_Y$, the matrix functions are evaluated in different subspaces, and their product may be less stable than the individual factors suggest. This coordinate mismatch does not necessarily cause large errors, but it removes the algebraic simplification that would otherwise stabilize the transport map.

An alternative is the *shared-basis* variant. Instead of computing two separate range finders, one constructs a single orthonormal basis Q that approximately spans the union of the dominant ranges of C_X and C_Y. A natural construction is to form the combined sketch matrix

$$Y_{\text{joint}} = [C_X\Omega,\ C_Y\Omega]$$

and compute Q as an orthonormal basis for the range of Y_{joint}. Both covariance operators are then projected onto this shared basis:

$$C_X \approx QB_XQ^\top, \qquad C_Y \approx QB_YQ^\top,$$

with $B_X = Q^\top C_X Q$ and $B_Y = Q^\top C_Y Q$. The regularized transport map becomes

$$\widehat{T}_\lambda = (QB_XQ^\top + \lambda I)^{-1/2}(QB_YQ^\top + \lambda I)^{1/2},$$

and both matrix functions are now evaluated in the same reduced coordinate system. This eliminates the coordinate mismatch and is expected to reduce $\|T_\lambda - \widehat{T}_\lambda\|_2$ even when the raw covariance errors $\|C_X - \widehat{C}_X\|_2$ and $\|C_Y - \widehat{C}_Y\|_2$ are comparable to those of the independent construction.

The shared-basis variant should not be understood as universally superior. Its advantage is a hypothesis to be tested experimentally: in regimes where the low-rank approximation is already reasonably accurate, shared coordinates may stabilize the transport map enough to make precision effects visible. In regimes where the approximation itself is too coarse, neither coordinate alignment nor precision choice will rescue the method. The experiments in ??→p.?? test this hypothesis explicitly.

1.5 Mixed Precision and Balancing Inexactness

This is the place where the climate application and the mixed-precision chapter meet most naturally. The important question is not whether lower precision is faster in the abstract. It is whether reduced precision changes the dependence correction less than the inexactness already present in the modeling pipeline. This is exactly the viewpoint advocated in recent mixed-precision work on balancing multiple sources of inexactness [9] [10].

For the proposed method, the natural arithmetic split is the following. The bulk products

$$Z\Omega, \qquad Z^\top(Z\Omega)$$

are the most plausible candidates for reduced precision. By contrast, the orthogonalization that produces Q, the small eigendecompositions of B_X and B_Y, and the evaluation of matrix square roots and inverse square roots are more sensitive and are therefore better kept in double precision. This follows the general mixed-precision pattern already visible in least-squares sketching and refinement methods: use cheaper arithmetic where the computation is large and structurally robust, and keep higher precision where instability would be amplified by conditioning [11] [12].

Accordingly, the computed covariance approximation should be written as

$$\widehat{C}_X = C_X + E_{\text{rand},X} + E_{\text{fp},X}, \qquad \widehat{C}_Y = C_Y + E_{\text{rand},Y} + E_{\text{fp},Y},$$

where E_{rand} denotes randomized approximation error and E_{fp} floating-point error. A concrete floating-point model enters already at the sample matrix level. If the dominant product is formed in lower precision with unit roundoff u_ℓ, then one may write

$$\widehat{Y}_X = \mathrm{fl}_{u_\ell}\Big(\frac{1}{n}\, Z_X^\top (Z_X \Omega)\Big) = C_X \Omega + \Delta Y_X,$$

and analogously for $\widehat{Y}_Y$. Under the standard matrix-product rounding model,

$$\|\Delta Y_X\|_2 \lesssim \gamma_n^{(u_\ell)}\ \frac{\|Z_X\|_2^2}{n}\ \|\Omega\|_2, \qquad \gamma_n^{(u_\ell)} = \frac{n u_\ell}{1 - n u_\ell}\,,$$

with the same form for ΔY_Y [10]. Thus the floating-point contribution depends not only on the unit roundoff, but also on the accumulation length, the scale of the latent data, the norm of the sketching operator, and on whether accumulation is performed in low or high precision. The key regime is the balanced one,

$$\|E_{\mathrm{fp}}\| \lesssim \|E_{\mathrm{rand}}\|,$$

or more generally when the floating-point perturbation remains below the statistical or modeling tolerance of the application. In that regime, computing the sketch in full double precision would amount to solving one part of the numerical problem much more accurately than the rest of the pipeline warrants.

Proposition 1.3 (Covariance sketch product perturbation). Let $Z \in \mathbb{R}^{n\times p}$ be a centered data matrix with empirical covariance $C = \frac{1}{n} Z^\top Z$, let $\Omega \in \mathbb{R}^{p\times \ell}$ be a sketching matrix, and let the computed sample product be formed in working precision with unit roundoff u. Write

$$\widehat{Y} = \mathrm{fl}_u\Big(\frac{1}{n}\, Z^\top (Z\Omega)\Big) = C\Omega + \Delta Y.$$

Then, under the standard inner-product rounding model with accumulation in precision u,

$$\|\Delta Y\|_2 \le \frac{nu}{1-nu}\ \frac{\|Z\|_2^2}{n}\ \|\Omega\|_2,$$

provided that $nu < 1$. In particular, writing $\eta_{\mathrm{fp}} = \|\Delta Y\|_2 / \|\Omega\|_2$, one obtains the operator-norm estimate

$$\eta_{\mathrm{fp}} \lesssim u\, \|Z\|_2^2 = u\, n\, \|C\|_2 = u\, n\, \lambda_{\max}(C),$$

so that the floating-point perturbation scales with the unit roundoff, the sample size entering the accumulation, and the spectral scale of the empirical covariance.

Remark. This bound models the combined product $Z^\top(Z\Omega)/n$ as a single inner-product accumulation of length n. In practice, the computation involves two stages — first $W = Z\Omega$ (accumulation over p terms per entry), then $Y = Z^\top W / n$ (accumulation over n terms per entry) — each with its own rounding contribution. A fully detailed two-stage bound would involve separate terms $\gamma_p^{(u)}$ and $\gamma_n^{(u)}$ for the respective accumulations. The present bound is a simplified first-order model that subsumes both stages into the dominant γ_n factor; the implementation provides an optional two-stage variant in `randMBC::sketch_forward_bound` with `two_stage = TRUE`.

Proof. This is a direct application of the standard matrix-product rounding model Thm. 3.5 [10]. Each column of $\frac{1}{n} Z^\top (Z\Omega)$ is an inner product of length n, so the columnwise relative perturbation is bounded by $\gamma_n = nu/(1 - nu)$. Taking norms and using $\|C\|_2 = \|Z\|_2^2/n$ gives the result. □

The practical content of Proposition 1.3→p.7 is that the floating-point contribution to the covariance sketch perturbation is computable from the data norms, the accumulation length, and the unit roundoff alone. The balanced regime

$$\eta_{\mathrm{fp}} \lesssim \eta_{\mathrm{rand}}$$

can therefore be tested before committing to a precision choice, by comparing η_{fp} against a randomized truncation error estimate (e.g. from a range-finder residual).

1.6 Perturbation Perspective

The central analytical task is not to study the entire bias-correction workflow at once, but to bound the perturbation of the transport map induced by the covariance approximation. Define

$$A = C_X + \lambda I, \qquad B = C_Y + \lambda I,$$

and let

$$\widehat{A} = A + \Delta A, \qquad \widehat{B} = B + \Delta B$$

be the regularized approximations produced by randomized low-rank compression and finite-precision arithmetic. Then

$$T_\lambda = A^{-1/2} B^{1/2}, \qquad \widehat{T}_\lambda = \widehat{A}^{-1/2} \widehat{B}^{1/2}.$$

The natural forward quantities are

$$\|T_\lambda - \widehat{T}_\lambda\|_2 \qquad \text{and} \qquad \|Z_f(T_\lambda - \widehat{T}_\lambda)\|_F.$$

At a first formal level one may split the transport error as

$$\|T_\lambda - \widehat{T}_\lambda\|_2 \le \|A^{-1/2}\|_2 \, \|B^{1/2} - \widehat{B}^{1/2}\|_2 + \|A^{-1/2} - \widehat{A}^{-1/2}\|_2 \, \|\widehat{B}^{1/2}\|_2.$$

This identity already reveals the main conditioning mechanism. Small covariance perturbations are not enough by themselves. Their effect is filtered through matrix square roots and, more severely, inverse square roots. Therefore the decisive quantity is not only the size of ΔA and ΔB, but also the spectral safety margin created by the regularization parameter λ.

If

$$\|\Delta A\|_2 \le \eta_A, \qquad \|\Delta B\|_2 \le \eta_B,$$

the later perturbation analysis can be organized into a small chain of deterministic statements.

Proposition 1.4 (Positive definiteness under perturbation). Let $A = C_X + \lambda I$ and $\widehat{A} = A + \Delta A$, where A and $\widehat{A}$ are symmetric. If $\|\Delta A\|_2 < \lambda_{\min}(A)$, then $\widehat{A}$ remains symmetric positive definite and

$$\lambda_{\min}(\widehat{A}) \ge \lambda_{\min}(A) - \|\Delta A\|_2.$$

The same statement holds for B and $\widehat{B}$.

Proof (Proof sketch). For any unit vector x,

$$x^\top \widehat{A} x = x^\top A x + x^\top \Delta A x \ge \lambda_{\min}(A) - \|\Delta A\|_2.$$

Taking the infimum over $\|x\|_2 = 1$ gives the bound. □

Proposition 1.5 (Perturbation of regularized square roots). Assume that A, $\widehat{A}$, B, and $\widehat{B}$ are symmetric positive definite, and set

$$\alpha_0 = \min\{\lambda_{\min}(A), \lambda_{\min}(\widehat{A})\}, \qquad \beta_0 = \min\{\lambda_{\min}(B), \lambda_{\min}(\widehat{B})\}.$$

Then

$$\|B^{1/2} - \widehat{B}^{1/2}\|_2 \le \frac{1}{2\beta_0^{1/2}} \|\Delta B\|_2,$$

and

$$\|A^{-1/2} - \widehat{A}^{-1/2}\|_2 \le \frac{1}{2\alpha_0^{3/2}} \|\Delta A\|_2.$$

Proof (Proof sketch). These bounds follow from the Fréchet-derivative structure of matrix square-root and inverse-square-root functions on the positive definite cone. For the square root, the Fréchet derivative $L_f(A, E)$ of the map $f(A) = A^{1/2}$ satisfies the Sylvester equation $A^{1/2} L_f + L_f A^{1/2} = E$; bounding $\|L_f\|_2$ via the spectral norm of the Sylvester operator yields the first inequality with the constant $(2\beta_0^{1/2})^{-1}$. The inverse-square-root bound then follows from the identity

$$A^{-1/2} - \widehat{A}^{-1/2} = A^{-1/2}(\widehat{A}^{1/2} - A^{1/2})\widehat{A}^{-1/2},$$

This algebraic identity is useful but does not by itself constitute a perturbation bound; the rigorous bound follows from matrix-function perturbation theory Chs. 3 and 6 [13]. Combined with the square-root perturbation bound and the norm estimates $\|A^{-1/2}\|_2 \le \alpha_0^{-1/2}$, $\|\widehat{A}^{-1/2}\|_2 \le \alpha_0^{-1/2}$. For a rigorous treatment of matrix-function Fréchet derivatives and perturbation bounds, see Chs. 3 and 6 [13]. The important qualitative point is the scaling: the inverse square root is more sensitive near small eigenvalues, with amplification proportional to $\alpha_0^{-3/2}$. □

Theorem 1.6 (Perturbation of covariance transport — first-order schematic bound)**.** Assume that the hypotheses of the previous proposition hold. Then

$$\|T_\lambda - \widehat{T}_\lambda\|_2 \le \|A^{-1/2}\|_2 \, \|B^{1/2} - \widehat{B}^{1/2}\|_2 + \|A^{-1/2} - \widehat{A}^{-1/2}\|_2 \, \|\widehat{B}^{1/2}\|_2,$$

and therefore

$$\|T_\lambda - \widehat{T}_\lambda\|_2 \le \frac{\|A^{-1/2}\|_2}{2\beta_0^{1/2}} \, \eta_B + \frac{\|\widehat{B}^{1/2}\|_2}{2\alpha_0^{3/2}} \, \eta_A.$$

In particular, the amplification factor may be taken as

$$K_\lambda(C_X, C_Y, \Delta A, \Delta B) = \max\left\{ \frac{\|A^{-1/2}\|_2}{2\beta_0^{1/2}} \, , \, \frac{\|\widehat{B}^{1/2}\|_2}{2\alpha_0^{3/2}} \right\},$$

so that

$$\|T_\lambda - \widehat{T}_\lambda\|_2 \le K_\lambda \, (\eta_A + \eta_B).$$

This is a first-order schematic bound used to guide experimental design. The constant in this bound is a schematic first-order amplification factor; it has not been tracked to a fully rigorous constant under the stated hypotheses. See the implementation's two-stage forward model for the numerically active version. When $\|\Delta A\|_2$ and $\|\Delta B\|_2$ are small relative to the regularized spectral gaps, the amplification factor is essentially a constant depending on $\lambda_{\min}(C_X + \lambda I)$ and $\lambda_{\min}(C_Y + \lambda I)$. In particular, the inverse-square-root term deteriorates at the rate $\lambda_{\min}(C_X + \lambda I)^{-3/2}$. A fully rigorous bound with sharper constants would require tracking higher-order terms and exact Sylvester-operator norms; the present statement suffices to explain the qualitative regime structure observed in the experiments.

Proof. Split the difference as

$$T_\lambda - \widehat{T}_\lambda = A^{-1/2}(B^{1/2} - \widehat{B}^{1/2}) + (A^{-1/2} - \widehat{A}^{-1/2})\widehat{B}^{1/2},$$

then apply the triangle inequality and the previous proposition. The final bound is first order in η_A and η_B, with constants determined by the regularized spectral gaps. □

The randomized and arithmetic contributions can then be separated as

$$\eta_A \le \eta_{\text{rand},A} + \eta_{\text{fp},A}, \qquad \eta_B \le \eta_{\text{rand},B} + \eta_{\text{fp},B}.$$

For spectra with visible decay, the randomized term is governed by the tail beyond the chosen rank, up to oversampling constants from the range-finder bound above [7] [8]. The floating-point term, by contrast, follows the sketch-product model above and is controlled by the working precision, the accumulation length, the scale of the latent data, and the norm of the sketching operator. The thesis question is precisely how these two error sources should be balanced against each other.

Algorithm 1: Mixed-precision randomized covariance transport

Input: Historical model data X_h, future model data X_f, historical reference data Y_h, target rank r, oversampling q, regularization λ, sketch precision u_ℓ

Output: Corrected future sample $\widetilde{X}_f^{\mathrm{corr}}$, diagnostics

Apply QDM-type marginal correction to obtain $\widetilde{X}_h$, $\widetilde{X}_f$;

Transform $\widetilde{X}_h$, $\widetilde{X}_f$, Y_h to latent rank-normal variables Z_X, Z_f, Z_Y;

`// Low-precision block:`;

Form randomized covariance sketches in precision u_ℓ (simulated or hardware-native): $Y_X = \frac{1}{n} Z_X^\top (Z_X \Omega_X), \quad Y_Y = \frac{1}{n} Z_Y^\top (Z_Y \Omega_Y)$;

`// High-precision block:`;

Compute orthonormal bases Q_X, Q_Y (or shared basis Q) via QR in double precision;

Form small covariance cores $B_X = Q_X^\top C_X Q_X$, $B_Y = Q_Y^\top C_Y Q_Y$ in double precision;

Build regularized low-rank covariance surrogates $\widehat{C}_{X,\lambda}$, $\widehat{C}_{Y,\lambda}$;

Evaluate regularized transport map in double precision: $\widehat{T}_\lambda = (\widehat{C}_{X,\lambda})^{-1/2} (\widehat{C}_{Y,\lambda})^{1/2}$;

Apply $Z_f^{\mathrm{corr}} = Z_f \widehat{T}_\lambda$;

Restore QDM marginals by rank rearrangement;

return corrected future sample and diagnostics;

Diagnostics: covariance error $\|C - \widehat{C}\|_2$; transport-map error $\|T_\lambda - \widehat{T}_\lambda\|_2$; induced future-data perturbation $\|Z_f T_\lambda - Z_f \widehat{T}_\lambda\|_F$; ratio $\eta_{\mathrm{fp}}/\eta_{\mathrm{rand}}$; runtime;

Corollary 1.7 (Perturbation of corrected future data)**.** Under the hypotheses of the theorem,

$$\|Z_f T_\lambda - Z_f \widehat{T}_\lambda\|_F \le \|Z_f\|_F \, \|T_\lambda - \widehat{T}_\lambda\|_2.$$

Proof. This is the submultiplicativity of the Frobenius norm with the operator norm on the right. □

1.7 Algorithmic Consequences

This formulation leads to a concrete algorithmic decomposition, presented formally as 1 →p.11.

This is the natural architecture for the software contribution as well. The implementation should separate a randomized covariance-approximation layer, a transport-map layer, and a climate-facing wrapper that handles marginal correction and rank restoration. Such a decomposition keeps the numerical core reusable while preserving a clear application entry point.

The local `randMBC` prototype used for the experiments follows this decomposition explicitly. It provides a deterministic marginal-correction and rank-restoration wrapper around a randomized covariance-transport core, together with a synthetic benchmark that compares the full regularized transport baseline, selected reference methods from the vendored `MBC` package, and the randomized low-rank implementation. The first controlled runs are informative even though they are not yet favorable to the randomized method: on the present synthetic benchmark, the deterministic full transport and the MBC references remain markedly more accurate than the low-rank randomized surrogate. This should not be read as a failure of the covariance-transport viewpoint itself. Rather, it identifies more precisely where the open problem lies, namely in the approximation regime where low-rank compression becomes accurate enough that transport-map perturbations no longer dominate the downstream dependence correction.

The same decomposition also determines the experimental program. The decisive sweeps are not arbitrary. They should vary the approximation rank, the regularization parameter, and the arithmetic precision while tracking covariance error, transport-map error, and the induced dependence error on the corrected future data. In particular, one should expect a regime where single precision is essentially indistinguishable from double because randomized truncation already dominates, and another regime where further rank increase reveals floating-point error because the approximation error has become too small to hide it.

1.8 Role in the Thesis

The purpose of this chapter is not to claim that covariance transport solves the whole problem of multivariate climate bias correction. It does not replace the need for application-side diagnostics, and it should not be presented as a universal substitute for richer multivariate schemes such as MBCn. Its role is narrower and more precise. It identifies a mathematically serious dependence-correction problem that is simultaneously

1. motivated by climate-model post-processing,
2. expressible in the language of randomized low-rank approximation,
3. sensitive to conditioning and regularization,
4. and naturally suited to mixed-precision analysis.

For the later parts of the thesis, this chapter therefore acts as a bridge. Upstream, it uses the perturbation viewpoint of ??→p.?? and the randomized approximation ideas from ??→p.??. Downstream, it provides the specific method family that the implementation and the climate-oriented experiments should test. In this sense, randomized covariance transport is the point at which the thesis becomes simultaneously application-aware and still unmistakably about numerical linear algebra.

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